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ARTICLE

Risk scores and reliability of the SARA, SARA-V3, B-SAFER, and ODARA among Intimate Partner Violence (IPV) cases referred for threat assessment

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ABSTRACT

Threat assessment services help police by identifying risk and recommending ways to mitigate risk for cases perceived to have a high potential for intimate partner violence (IPV). However, research has yet to show that cases referred for threat assessment score higher on IPV risk tools than routine policing samples, which would show whether referrals are appropriate. Furthermore, it is unknown whether these tools can be scored reliably from documents gathered by threat assessors without contacting perpetrators or victims. We scored the Spousal Assault Risk Assessment Guide (SARA) and its revision (SARA-V3), Brief Spousal Assault Risk Evaluation Form (B-SAFER), and Ontario Domestic Assault Risk Assessment (ODARA) from threat assessment files of 238 men referred for IPV offenses. Cases scored higher than previously reported routine policing samples. Inter-rater reliability coefficients for total scores and most subscales were ≥.70. Findings support appropriateness of referrals and capacity to reliably score IPV risk tools.

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Violence against an adult marital, cohabiting, or dating partner (i.e., intimate partner violence, IPV) places a substantial strain on policing services. IPV represents the largest single category of calls to police in the United States (e.g., Klein, 2009). In Canada, more than one in four police-reported victims of violence are intimate partners of the person who perpetrated the offense (e.g., Burczycka & Conroy, 2017). Policing services are increasingly using structured risk assessments developed to aid decision making from the point of arrest (e.g., Storey et al., 2014; Hilton & Eke, 2017). Police appear more likely to arrest individuals who pose measurably higher risk of IPV recidivism (e.g., Hilton et al., 2007), and large-scale studies suggest that arrest may reduce recidivism for most offenders (e.g., Johnson & Goodlin-Fahncke, 2015; Maxwell et al., 2002). Furthermore, when police officers assess cases to be at high risk of repeated IPV, they make more recommendations for managing risk, including referring cases for further risk assessment (Belfrage et al., 2012; Storey et al., 2014). In turn, threat assessment services are largely occupied with cases involving IPV (e.g., Ennis et al., 2015; Kropp & Cook, 2014; Storey et al., 2014). The present study examined whether

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IPV cases referred to a police threat assessment service following initial assessment by investigating officers score higher than routine cases on IPV risk assessment measures, and whether these measures can be reliably scored from the information available to threat assessors.

Determining whether cases referred for threat assessment are higher risk than the general population of IPV arrestees represents an important first step in evaluating the utility and effectiveness of threat assessment services, and in particular the appropriateness of referrals. The principles of effective correctional service state that the highest risk cases should receive the most intensive response in terms of custody, supervision, or other interventions (e.g., Bonta & Andrews, 2017). Importantly, services that adhere to the risk principle have a significantly larger effect on reducing recidivism than those that do not (e.g., Lowenkamp et al., 2006). In contrast, violating the principles of effective correctional services has the potential to increase criminal recidivism (e.g., Barnett & Howard, 2018). Concomitant social and individual harms may include increased victimization, unfair detention for lower risk individuals, and inefficient use of criminal justice resources.

The application of the risk principle to IPV cases referred for threat assessment has not been previously studied, despite the intensity of service provided by threat assessors. Threat assessors are highly qualified personnel; they typically include senior police officers, criminal intelligence analysts, and other trained civilians with criminal justice education backgrounds, as well as psychologists and psychiatrists in some settings. Threat assessors conduct comprehensive risk assessments, consider factors that potentially increase or decrease risk, formulate scenarios under which future violence might occur, and make case management recommendations intended to help prevent future violence (e.g., Ennis et al., 2015). The intensity of resource provision is justified on the basis that threat assessors provide needed support to regional police in cases initially considered to be high risk. However, despite IPV cases predominating in threat assessment caseloads (e.g., Ennis et al., 2015; Kropp & Cook, 2014; Storey et al., 2014), no research to date has tested whether individuals referred for IPV threat assessment score higher on IPV risk assessment tools than the overall population of men apprehended by police for IPV.

Another distinct aspect of threat assessment services is that they are intended to review and assess information that has been gathered during the police investigation, along with supplementary records where available, such as probation and court records, psychological assessments, and child protection services documents. They do not contact the persons being assessed, interview victims, or conduct mental health assessments. As a result, the information that threat assessors use may lack some information that is key to scoring some risk assessment items, in contrast to other assessment settings where this information may be more easily accessed over time, such as in correctional or community mental health settings. Therefore, there is a pressing need to ensure that the information gathered in the threat assessment context is sufficient for reliable scoring of IPV risk assessment tools.

IPV risk assessment tools are widely available and several have been validated in frontline policing contexts (e.g., Graham et al., 2019; Svalin & Levander, 2020), so it is now feasible to address these knowledge gaps. In the present study, therefore, we compared the risk scores for a threat assessment sample to those previously reported for routine police cases of IPV, and tested inter-rater and internal reliability in a sub-sample of threat assessment cases.

IPV risk assessment

Prior to the development and validation of structured guides to the assessment of risk, assessors relied entirely on unstructured professional judgement. Meta-analysis revealed that mechanical and statistical methods outperform clinical judgement of violence risk (e.g., Grove et al., 2000; Hilton et al., 2006). Several well-established tools are now available for measuring the risk of IPV recidivism. These tools are typically validated by testing their ability to discriminate between individuals who do and do not reoffend, in follow-up studies of those who are accused or adjudicated for IPV offenses. Discriminative validity can be reported using effect sizes, such as the receiver operating characteristic area under the curve (AUC) statistic. AUCs range from 0 to 1, where .5 indicates chance association between the predictor and the outcome, AUCs over .56, .64, and .71 are considered small, medium, and large effects predicting the outcome, respectively (Rice & Harris, 2005), and AUCs below .5 indicate prediction in the opposite direction.

Graham et al. (2019) identified 17 IPV risk assessment tools whose psychometric properties had been examined in at least one study. The two most studied tools for assessing IPV recidivism risk are the Ontario Domestic Assault Risk Assessment (ODARA; Hilton, Harris, & Rice, 2010; Hilton et al., 2004) and the Spousal Assault Risk Assessment Guide (SARA; e.g., Kropp et al., 1995; Kropp & Hart, 2000). The present study focuses on the ODARA and the SARA family of tools.

The SARA was one of the first published tools for IPV risk assessment. The SARA was originally developed for use in threat assessment contexts (Cook et al., 2014). This tool uses the structured professional judgment approach, in which items are rationally derived from the empirical or theoretical literature related to IPV outcomes. The SARA contains 20 items concerning the psychosocial history, IPV and general criminal history of the person being assessed, and aspects of the most recent IPV offense; items have shown good internal consistency ($\alpha = .78$; Kropp & Hart, 2000). The tool yields a total item score that is used, in conjunction with other information that the assessor identifies as important, to make a categorical judgment of risk. Predictive validity studies have used the SARA score more often than the categorical judgment, but both total scores and judgments predicted IPV recidivism with medium effect sizes in a review of published and unpublished studies with samples drawn from police, corrections, and court-mandated treatment settings (average AUCs = .63 to .67; Helmus & Bourgon, 2011; see also Messing & Thaller, 2013). A more recent review by Graham et al. (2019) reported that AUCs for SARA scores and judgments range from .52 to .72.

The SARA Version 3 (SARA-V3; Kropp & Hart, 2015) also has 20 items, some of which overlap with the SARA items. The SARA-V3, however, adds items pertaining to victim vulnerability. Evaluation of items on the SARA-V3 is subdivided into consideration of events that occurred in the past (i.e., greater than 1 year prior to the assessment date) and events that occurred recently (i.e., within 1 year of the assessment date). The SARA-V3 encourages assessors to consider the influence of preselected items and other factors potentially relevant to risk in an individual case, and then formulate scenarios in which violence could occur, along with possible risk management strategies, before forming an opinion about the risk of future IPV. We are not aware of any published validations of the SARA-V3's predictive accuracy to date. However, we included it in the present study because it is especially intended for the threat assessment professional rather than frontline policing.

The Brief Spousal Assault Form for the Evaluation of Risk (B-SAFER; Kropp et al., 2010; Kropp & Hart, 2004) is a condensed, 15-item version of the SARA designed for use by police officers responding to IPV incidents. As with the SARA, assessors are to score the B-SAFER items, which then inform a summary rating of risk. Like the SARA-V3, B-SAFER items are separated into *past* (over 4 weeks prior to assessment) and *recent* (during the 4 weeks prior to assessment) events. Graham et al. (2019) report on one validation study of the B-SAFER with an AUC of .70. B-SAFER total scores predicted criminal recidivism among 40 men imprisoned for a violent or other offense against an intimate partner (AUC = .76; Loinaz, 2014), but did not significantly predict IPV-specific recidivism among 158 men court-ordered to treatment for IPV violence or stalking (AUC = .55, Gerbrandij et al., 2018). Similarly, mixed findings have been reported for the B-SAFER summary rating. That is, in one study, both total scores and the summary rating predicted new IPV assaults among 249 men with a police record of IPV (AUC = .70 for scores, AUC = .69 for ratings; Storey et al., 2014). However, Belfrage and Strand (2012) reported that police officers' summary ratings were not significantly related to new assaults against an intimate partner among 216 men (test statistics were not reported). Svalin et al. (2018) tested summary ratings of likelihood and severity for their association with several outcomes, in 301 assessments (including 11 cases of female-perpetrated offenses) and reported only one significant effect.

The Ontario Domestic Assault Risk Assessment (ODARA; Hilton et al., 2004) was also designed for use by frontline police. The ODARA has 13 items that cover the most recent IPV incident, the police and criminal record and other antisocial behavior of the individual being assessed, and children in the relationship. The ODARA has shown good internal consistency (α = .65; Hilton et al., 2008). The ODARA is an actuarial tool, in that its items were selected from follow-up research identifying the strongest predictors of recidivism, and individuals' risk of IPV recidivism is interpreted from actuarial experience tables using total scores. The tool was constructed in a sample of 589 men with a police record for IPV against a female marital or cohabiting partner, and then cross-validated in similar samples with moderate to large effects (AUCs = .65 to .80; Hilton & Harris, 2009; Hilton et al., 2008, 2004). In other studies, ODARA scores predicted recidivism by men with a police record of IPV (e.g., Gerth et al., 2015; Jung & Buro, 2017; Hilton & Eke, 2016), in a combined sample of men and women with a police record of IPV (Olver & Jung, 2017), and by men in correctional or forensic settings (e.g., Rettenberger & Eher, 2013; Seewald et al., 2017; Hilton et al., 2010). Messing and Thaller (2013) reported an average AUC of .67 for published validation studies of the ODARA, and Graham et al. (2019) reported a range of .61 to .86.

Inter-rater reliability

Risk management relies on dependable measures of risk, and reliability is a prerequisite of validity (e.g., Anastasi & Urbina, 1997). Reliability can refer to internal consistency measured by item-to-total correlations, or to level of agreement between raters. Inter-rater reliability is typically assessed with intra-class correlation (ICC) coefficients, rather than Pearson *r* which measures simple association of scores. Both range from a high of 1 (perfect agreement) through 0 (no consistency) to −1 (perfect disagreement).

SARA scores have shown inter-rater reliability coefficients of .84 and higher among case managers, psychologists, and students (Grann & Wedin, 2002; Kropp & Hart, 2000; all studies reported here used the ICC except where noted). Reliability of .63 was reported for the SARA summary ratings (Kropp & Hart, 2000). Reliability was excellent for B-SAFER total scores (.96; Gerbrandij et al., 2018); agreement for B-SAFER risk judgments ranged from .63 to .88 in cases assessed by clinicians (Gerbrandij et al., 2018) and .58 to .72 among police officers (Svalin et al., 2017, using Kendall's τ for ranked measures). ODARA score reliability has been reported to be .80 to .95 between researchers (Hilton et al., 2004; Hilton, Harris, Popham, et al., 2010; Rettenberger & Eher, 2013; see also Hilton & Eke, 2016, who reported Pearson correlations ≥ .80) and .95 between officers (Hilton et al., 2004).

Where a risk assessment tool is interpreted based on the overall result, reliability of the final score or judgment is more important than agreement on individual items. However, item reliability is also desirable in threat assessment because some IPV tools include individual items intended to help determine specific risk management strategies. From a structured professional judgment perspective, it has been argued that, 'If raters cannot agree on the presence of individual risk factors … there is little point in conducting risk assessments' (Kropp & Hart, 2000, p. 109). Reliability of SARA items has ranged from .30 to 1.00 and averaged .58 to .65 (Grann & Wedin, 2002; Kropp & Hart, 2000). Four B-SAFER items scored by probation officers had an average coefficient of .57 (Thijssen & de Ruiter, 2011) and coefficients for the 15 items ranged from −.34 to .71 among police officers (Svalin et al., 2017, using τ). Jung and Buro (2017) tested reliability between researchers and found coefficients from .12 to .80 for SARA items and .36 to 1.0 for ODARA items, using the more conservative kappa correlation coefficient that adjusts for the chance rate of agreement.

As correlations can be reduced in samples with limited variability, percentage agreement can be informative for items with extremely high or low prevalence. Jung and Buro (2017) reported that percentage agreement on SARA items ranged from 40% to 94%, and ODARA items from 74% to 100%. Svalin et al. (2017) reported percentage agreement on B-SAFER items scored by police officers, ranging from 22% to 96%. Inter-rater reliability tests for individual items from IPV tools have been limited compared with studies of total scores and, to date, no inter-rater reliability tests have been conducted in a threat assessment sample.

The present study

We examined the psychometric properties of the SARA, SARA-V3, B-SAFER, and ODARA as scored from police and criminal records in a sample of men charged with IPV against their female intimate partners and referred for threat assessment. We compared the present sample's risk assessment scores with those of previously published samples of men drawn from routine police occurrence reports of IPV, expecting the present sample to be relatively high risk. We then examined correlations among risk scores from the four measures and each measure's internal consistency, and tested inter-rater reliability for each tool and for the items within each tool. This study is the first investigation of the reliability of risk assessment when scored from the police and criminal records of men referred to threat assessment for IPV.

Method

The research protocol was reviewed and approved by the research ethics boards of four authors' institutions, and a formal notice of collaboration was obtained from the law enforcement agency where the study took place. All data were extracted by research coders from files containing threat assessor reports and supporting documentation pertaining to cases assessed in 2010 through 2015.

Context

All data were gathered from a threat assessment service that provided assessment and consultation services to police and other government agencies throughout one Canadian province. The threat assessment service was a multi-agency, joint forces unit, comprising seconded members from police services throughout the province. Prior to 2016, there was no protocol guiding which cases should be referred for threat assessment, but the threat assessment service encouraged police investigators to refer any cases that caused them significant concern regarding serious or recurrent IPV. At the time, structured risk assessment for IPV was not standard practice in the referring police services, so investigators typically relied on their professional discretion or informal procedures within their units to determine which files to refer. Once a case was referred, officers at the local police service would continue their investigation while awaiting the assessment report provided by the threat assessment service.

Samples

There were 258 men charged with a violent offense against a female intimate partner whose files were referred for threat assessment between 2010 (when the threat assessment service began using validated IPV risk assessment tools) and 2015 (just prior to the start of data collection). We excluded 20 cases for whom information was lacking to code all risk assessment tool items, bringing the final sample to 238 for the current study. The men's mean age at the time of the index offense was 35 years (SD = 9.11, range 18 to 65). Most were of White/European descent (n = 160, 67%), and others were identified as Indigenous (n = 56, 24%), or 'other' or missing ethnicity information (n = 21, 9%). Less than half were employed at the time of the offense (n = 100, 42%), and others were unemployed (n = 94, 39.5%) or missing case file information on employment status (n = 44, 18.5%).

The subset of 32 cases drawn at random for tests of inter-rater reliability had a mean age of 39 years (*SD* = 10.05). The subsample size was chosen on the basis of consistency with previous research, described above in our review of the literature on the inter-rater reliability of IPV risk assessment tools. We aimed to have at least 10% of the final sample. Significant inter-rater agreement has been reported with as few as eight test cases (Gerbrandij et al., 2018), and up to 86 have been used (Kropp & Hart, 2000), but most studies have used 10 to 30 test cases (Grann & Wedin, 2002; Hilton & Eke, 2016; Hilton et al., 2004; Hilton, Harris, Popham, et al., 2010; Jung & Buro, 2017; Rettenberger & Eher, 2013; Thijssen & de Ruiter, 2011). These men were predominantly of White/European descent (*n* = 25, 78%), Indigenous (*n* = 4, 13%) or 'other' (*n* = 3, 9%).

Data used to compare the psychometric properties of the threat assessment sample with routine police samples were as reported by Belfrage et al. (2012) for the SARA, and by Gerbrandij et al. (2018) for the 15-item version of the B-SAFER. For the ODARA, we compared our sample with the routine police samples reported by Hilton et al. (2004) and by Jung and Buro (2017). Jung and Buro's study was conducted in a municipality that is within the same jurisdiction as the current project. No published data were available for the SARA-V3.

Measures

SARA, SARA-V3, B-SAFER, and ODARA items were coded from records maintained by the threat assessment service. Threat assessors at this service use three main sources of information, which were also available to the researchers: (1) threat assessment request forms, which included information on the index offense and documented historical occurrences of IPV; (2) official federal and provincial criminal records and police occurrence reports, extracted at the time the threat assessment was conducted (police occurrence reports typically included investigating officers' handwritten notes and typed reports, documented evidence, and arrest details for each occurrence); and (3) a brief case history compiled by the threat assessors. Documents from other sources were also used where threat assessors had obtained them as part of their evaluation (e.g., probation and court records, psychological assessments, and child protection services documents). Thus, research coders had access to the same case file information that threat assessors used for their evaluation. Information on reoffending and other relevant behavior subsequent to the assessment date was unknown to coders.

We previously described in detail the four risk assessment tools and the evidence regarding their predictive validity and inter-rater reliability in the introduction. Here, we provide additional details of their items, how we coded them, and the average scores in routine police samples.

The SARA authors recommend using the tool as a general guideline to assess risk, rather than mechanically adding the item scores for a total risk score. Nevertheless, empirical studies have often used the total risk score in statistical analyses to investigate properties of the SARA and its relationship with other variables (e.g., Belfrage et al., 2012; for review, see Helmus & Bourgon, 2011). We coded each of the SARA's 20 items using the 3-point nominal scale, where '0' = *no or absent*, '1' = *possibly or partially present*, and '2' = *yes or present* (Kropp et al., 1995). As is commonly done in research, we also calculated a total risk score by summing the item scores. The SARA total score had a possible range of 0 to 40. In samples of men assessed by police, the mean SARA total score has ranged from 3.11 to 9.41 (Belfrage et al., 2012; Williams & Houghton, 2004; Hilton et al., 2004). Total scores were significantly higher among inmates (16.39) than probationers (12.98; Kropp & Hart, 2000).

We coded each of the SARA-V3's 40 items (Kropp & Hart, 2015) using the same 3-point nominal scale as for the SARA and summed item scores for a total score. The SARA-V3 total score had a possible range of 0 to 80; subtotal scores on the past and recent items both had a possible range of 0 to 40. We are aware of no reports of SARA-V3 scores in routine police samples.

We used a similar approach for coding the B-SAFER (Kropp et al., 2010; Kropp & Hart, 2004) items and total score. The B-SAFER total score had a possible range of 0 to 60; subtotal scores on the past and recent items both had a possible range of 0 to 30. Average B-SAFER total scores in policing samples have been reported in the range of 9.00 to 10.46 (Gerbrandij et al., 2018; Storey et al., 2014).

We coded the ODARA's 13 items according to the published scoring criteria (Hilton, Harris, & Rice, 2010). Each item was scored 1 = *present* or 0 = *absent*, and the total score was calculated as the sum of all items present. We report raw total ODARA scores, which have a possible range of 0 to 13. In routine policing samples, ODARA total scores have averaged from 1.9 to 5.7 (Jung & Buro, 2017; Hilton & Harris, 2009).

Procedure

All data were extracted from threat assessment case files by trained undergraduate and graduate research assistants from 2016 to 2018 and quantified using coding forms accompanied by a coding manual with detailed instructions. The training process involved two stages. First, coders received standard training on the risk assessment tools, including the ODARA online training at http://odara.waypointcentre.ca (described by Hilton & Ham, 2015) and training for the structured professional judgment guideline tools (SARA, SARA-V3, B-SAFER) online at https://proactive-resolutions.com. Then, coders were extensively trained on the coding instructions by a graduate research coordinator using two threat assessment cases previously coded by two of the research investigators. In addition, coders were tested for coding accuracy on at least one case also previously coded by primary investigators, before continuing to code independently for inter-rater reliability tests. Later, coders completed at least one subsequent test case previously coded by the research investigators, to ensure continued adherence to coding instructions and avoid coding drift or complacency.

Data analysis

We computed total scores, and subscale scores where applicable, for each of the IPV risk assessment tools using descriptive statistics (mean and standard deviation). We measured internal consistency for each scale and subscale using Cronbach's alpha, and intercorrelations among scores using Pearson correlation coefficients. Where previous studies of routine police samples reported means, standard deviations, and sample sizes for the IPV risk assessment tools, we compared them with the mean scores in the full sample using *t*-tests and Cohen's *d*.

We then tested inter-rater reliability for each total and subscale score in the sub-sample of 32 cases using the ICC absolute agreement, mixed model. We tested inter-rater reliability for the items within each tool using both the kappa coefficient and percent agreement.

Results

Table 1 shows the mean total scores on the SARA, SARA-V3, B-SAFER, and ODARA, and the past and recent sub-scores for the SARA-V3 and B-SAFER in the full sample. All measures had higher mean scores than reported in previous studies of IPV perpetrators drawn from police records, as described in the introduction. The mean SARA score was 22.87 (SD = 5.77) and is higher than the range reported for Belfrage et al.'s (2012) European police sample (M = 9.41, SD = 5.99, N = 429), $t\,(459) = 12.29$, $p < .001$, $d = 2.29$. The SARA-V3 mean total score was 48.31 (SD = 14.28) and Table 1 shows past and recent subscales of the SARA-V3 (no comparison available). The B-SAFER total score for our sample had a mean of 22.11 (SD = 6.25) and was higher than the range reported in Gerbrandij et al.'s (2018) United States study (M = 10.46; SD = 4.23, N = 158), $t\,(188) = 13.00$, $p < .001$, $d = 2.18$. The mean ODARA score for this sample fell into the highest actuarial risk category. The full sample had a mean ODARA total score of 7.84, SD = 2.03, which was significantly higher than the routine police sample reported in the original Canadian study (Hilton et al., 2004), M = 2.89, SD = 2.14, N = 589; $t\,(825) = 30.56$, $p < .001$, $d = 2.37$. Scores ranged from 2 to 12 compared with 0 to 11 in the

Table 1. Risk assessment scores, internal consistency, and intercorrelations among risk assessment scores in the whole sample (N = 238), and inter-rater reliability in the sub-sample (n = 32).

Measure	Mean (*SD*)	Inter-Rater Reliability	Intercorrelations among Risk Assessment Scores							
			1	2	3	4	5	6	7	8
1. SARA	22.87 (5.77)	.70***	(.66)	.76**	.63**	.59**	.59**	.68**	.15*	.60**
2. SARA-V3	48.31 (14.28)	.83***		(.87)	.87**	.73**	.70**	.80**	.17**	.48**
3. SARA-V3 Past	23.90 (10.21)	.86***			(.89)	.30**	.55**	.69**	.05	.50**
4. SARA-V3 Recent	24.41 (7.36)	.85***				(.78)	.60**	.59**	.26**	.24**
5. B-SAFER	22.11 (6.25)	.70***					(.73)	.79**	.67**	.40**
6. B-SAFER Past	19.68 (4.66)	.83***						(.68)	.07	.42**
7. B-SAFER Recent	2.43 (3.84)	.22							(.82)	.15*
8. ODARA	7.84 (2.03)	.84***								(.40)

Inter-rater reliability is measured by intra-class correlation coefficients (ICC) mixed model, absolute agreement, in the sub-sample of 32 cases. Internal reliability is measured by Cronbach's α shown on the diagonal in parentheses. Intercorrelations are measured by Pearson correlation coefficients (*r*), using the full sample. * p <.05, ** p <.01, *** p <.001.

Hilton et al. (2004) sample. It was also significantly higher than the routine police sample reported by Jung and Buro (2017) drawn from a city in the same Canadian province as the threat assessment service ($M = 5.7$, $SD = 2.7$, $N = 226$), $t\ (462) = 9.68$, $p < .001$, $d = 0.90$.

Scale reliability

Table 1 also shows intercorrelations among instrument scores in the full sample. All were positive and statistically significant, except for the B-SAFER recent score with the B-SAFER and SARA-V3 past scores. Cronbach's alpha coefficients (α, provided on the diagonal of Table 1) ranged from .66 to .89 for total and subscale scores of the SARA, SARA-V3, and B-SAFER. However, alpha was lower for the actuarial tool, ODARA (see Table 1).

Inter-rater reliability tests were conducted on scale totals and item scores in the sub-sample of 32 cases, as scored by research coders. Total and subscale score inter-rater reliability ranged from .70 to .86 (see Table 1), except for the B-SAFER recent subscale, ICC = .22.

For individual items, inter-rater reliabilities varied substantially. For SARA items, the mean calculable κ was .43, and agreement ranged from 31% to 94% (see Table 2). For SARA-V3 items,

Table 2. Inter-rater reliability of SARA version 2 item scores between two research coders.

SARA Item	κ	Percentage Agreement
1. Past assault of family members	.70***	84
2. Past assault of strangers or acquaintances	.69***	88
3. Past violation of conditional release or community supervision	.38**	66
4. Recent relationship problems	.72***	94
5. Recent employment problems	.38**	59
6. Victim of and/or witness to family violence as a child or adolescent	.59***	78
7. Recent substance abuse/dependence	.87***	94
8. Recent suicidal or homicidal ideation/intent	.15	44
9. Recent psychotic and/or manic symptoms	.13	78
10. Personality disorder with anger, impulsivity, or behavioral instability	−.03	31
11. Past physical assault	-	94
12. Past sexual assault/sexual jealousy	.30**	53
13. Past use of weapons and/or credible threats of death	.39**	75
14. Recent escalation in frequency or severity of assault	.33**	56
15. Past violation of 'no contact' orders	.51***	69
16. Extreme minimization or denial of spousal assault history	.27*	56
17. Attitudes that support or condone spousal assault	.41**	63
18. Severe and/or sexual assault	.51***	69
19. Use of weapons and/or credible threats of death	.72***	84
20. Violation of 'no contact' order	.65***	81

Reliability based on dichotomous item scoring (0 versus 1 or 2) as that is how threat assessors reported the SARA items. κ = kappa coefficient. *N* = 32. * p <.05, ** p <.01, *** p <.001.

Table 3. Inter-rater reliability of SARA version 3 item scores between two research coders.

	Past		Recent	
SARA-V3 Item	κ	Percentage Agreement	κ	Percentage Agreement
N1. Intimidation	.84***	91	.88***	94
N2. Threats	.75***	88	.40*	72
N3. Physical harm	.63***	88	.46***	88
N4. Sexual harm	.69***	84	.62***	81
N5. Severe IPV	.33**	63	.68***	84
N6. Chronic IPV	1.00***	100	1.00***	100
N7. Escalating IPV	.01	38	.30*	56
N8. IPV-related supervision violations	.71***	84	.50**	50
P1. Intimate relationships	.64***	88	.63***	91
P2. Non-intimate relationships	1.00***	100	1.00***	100
P3. Employment/finances	.83***	91	.70***	81
P4. Trauma/victimization	.56***	75	.03	69
P5. General antisocial conduct	.46***	69	.32**	63
P6. Major mental disorder	.38***	63	.47***	69
P7. Personality disorder	.05	41	.22	50
P8. Substance use	.72***	88	.94***	97
P9. Violent/suicidal ideation	.23**	47	.13	38
P10. Distorted thinking about IPV	.56***	72	.88***	94
V1. Barriers to security	.25*	69	.22	63
V2. Barriers to independence	.49***	81	.39**	75
V3. Interpersonal resources	.29*	88	.53**	88
V4. Community resources	–	97	–	97
V5. Attitudes or behavior	.44***	63	.24	50
V6. Mental health	.46***	78	.17	66

κ = kappa coefficient. N = 32. * p <.05, ** p <.01, *** p <.001

Table 4. Inter-rater reliability of B-SAFER item scores between two research coders.

	Past		Recent	
B-SAFER Item	κ	Percentage Agreement	κ	Percentage Agreement
1. Violent acts	–.02	94	–	94
2. Violent threats or thoughts	.32*	81	–.03	94
3. Escalation	.81***	94	–	97
4. Violation of court orders	.45**	78	–	97
5. Violent attitudes	.86***	72	1.00***	100
6. General criminality	.39**	63	–	100
7. Intimate relationship problems	–	97	–	100
8. Employment problems	.87***	94	.67***	88
9. Substance use problems	.67***	84	–	100
10. Mental health problems	.16	47	–.05	84
11. Inconsistent attitudes or behavior	.95***	97	.18	81
12. Extreme fear of perpetrator	.18	47	–	97
13. Inadequate support or resources	.48**	94	–	97
14. Unsafe living situation	.78***	94	–	94
15. Health problems	.62***	78	–	97

κ = kappa coefficient. N = 32. * p <.05, ** p <.01, *** p <.001

agreement ranged from 41% to 100% for past items (mean κ = .54) and 38% to 100% for recent items (mean κ = .51; see Table 3). For B-SAFER items, agreement ranged from 47% to 97% for past items (mean κ = .54) and 81% to 100% for recent items (mean κ = .45; see Table 4). For ODARA items, agreement ranged from 69% to 97% (mean κ = .62; see Table 5).

Discussion

The present study examined the risk scores of men referred for threat assessment following investigation for IPV, and psychometric properties of the risk assessment scales, using the SARA, SARA-V3, B-SAFER, and ODARA. Our results provide evidence that the threat assessment sample

Table 5. Inter-rater reliability of ODARA item scores between two research coders.

ODARA Item	κ	Percentage Agreement
Pre-index domestic assault	−.08	84
Pre-index non-domestic assault	.65***	84
Pre-index custodial sentence ≥ 30 days	.75***	88
Failure on pre-index conditional release	.54**	84
Threat to physically harm at index	.67***	81
Confinement of partner at index	.78***	88
Victim concern about future assaults	.43*	69
More than one child altogether	.46**	81
Victim's biological child previous partner	.93***	97
Pre-index non-domestic violence	.63**	88
Two indicators of substance abuse	.69***	88
Assault on index victim when pregnant	.80***	84
Barriers to victim support	.76***	94

κ = kappa coefficient. *N* = 32. * *p* <.05, ** *p* <.01, *** *p* <.001

posed a relatively high risk of IPV recidivism compared with published data drawn from routine police records of individuals apprehended for IPV offenses. Mean risk scores were considerably higher than comparable scores for the SARA and B-SAFER in previously published samples (e.g., Belfrage et al., 2012; Gerbrandij et al., 2018; Jung & Buro, 2017; Storey et al., 2014; Williams & Houghton, 2004). ODARA scores were in the highest actuarial risk category, on average, and significantly higher than the original policing sample (Hilton et al., 2004) and a routine sample drawn from the same province as the current threat assessment cases (Jung & Buro, 2017).

Notably, our effect sizes for mean score comparisons on the structured professional guideline tools were greater than 2, and greater than or equal to 0.9 for the actuarial tool, illustrating the large difference (Cohen, 1992) in risk between our sample and previous routine police samples. Given that effective correctional service involves tailoring the intensity of custody, supervision, and other interventions based on assessed levels of risk (e.g., Bonta & Andrews, 2017), our results suggest that investigating police officers are effectively identifying higher risk cases and appropriately referring them for intensive threat assessment service.

All scale total scores and subscale scores showed good inter-rater reliability, except the B-SAFER recent subscale. Reliability coefficients were mostly in the moderate and large range (Cohen, 1992), meeting or surpassing levels of agreement reported in previous research (e.g., Gerbrandij et al., 2018; Grann & Wedin, 2002; Kropp & Hart, 2000; Rettenberger & Eher, 2013; Thijssen & de Ruiter, 2011; Hilton & Eke, 2016; Hilton, Harris, Popham, et al., 2010; Hilton et al., 2004). This level of agreement is noteworthy given that our sample consisted of relatively high risk cases, which restricted the range of scores. Agreement on individual items was more varied, with stronger agreement for items pertaining to criminal and relational behavior (e.g., past assaults, use of weapons) and lower agreement for items relating to mental health (e.g., personality disorder, suicidal or homicidal ideation) and some victim-related variables (e.g., extreme fear of the person being assessed, mental health, access to community resources).

Previous studies of IPV risk assessment by police officers have found that items associated with mental health are among those most likely to be omitted or rated as absent (Belfrage et al., 2012; Loinaz, 2014; Kropp & Hart, 2004), a finding attributed to police officers' limited ability to identify mental disorder and evaluate its severity (Kropp & Hart, 2004). Jung and Buro (2017) also reported difficulties in reliably coding mental health variables from police files. Threat assessors were able to request additional collateral documentation from correctional facilities and mental health reports not normally available to investigating officers responding to a domestic occurrence. Nevertheless, our results, together with studies using routine police files, suggest that available records simply do not include sufficiently abundant or detailed mental health information to allow for consistent and accurate coding of mental health items.

A similar interpretation may apply to our finding of lower agreement on items concerning victims' mental health and other circumstances. This finding may be attributable in part to insufficient relevant information on the case files, as police investigators focus on characteristics of the person being assessed and record less information about victims. Some files did contain mental health information, but this was usually limited to comments made by the investigating officers rather than a full clinical report or diagnosis, and may have been difficult for research coders to interpret and code. It is possible that IPV risk assessment items to do with mental health are poorly suited for use in police and threat assessment contexts, or their operational definitions need adjusting to accommodate the limited mental health expertise of investigating officers and threat assessors. Ultimately, the extent to which limitations in the nature and quality of police files create problems for using existing IPV risk assessment tools that include mental health and victim vulnerability information depends on how important such information is for estimating risk and determining priorities for risk management. Research with correctional service files suggests that antisocial behavior and victim vulnerability factors weigh more heavily than nature of the index offense in the prediction of IPV recidivism (e.g., Hilton et al., 2008). This finding suggests that the resources needed to gather and evaluate mental health information would be worthwhile, especially for relatively high risk individuals such as those referred for threat assessment.

We found that internal consistency was higher for the structured professional tools (SARA, SARA-V3, and B-SAFER) than the actuarial tool (ODARA). Although the ODARA's items exhibited lower inter-correlations, this finding does not necessarily imply that the ODARA has a more complex factor structure than the other tools (e.g., Clark & Watson, 2019). A better understanding of the dimensionality of IPV risk measures and related concepts of IPV risk could be established in future research using factor analytic techniques.

Limitations and future research

Inter-rater reliability is a property of a particular instance of scoring the tool and not an immutable characteristic of the tool itself; therefore, limitations of the scoring procedures in the current study must be noted. Researchers scored the risk assessment measures from information that was available in archival threat assessment reports, which were limited especially with respect to mental health and victim-related information, for reasons described above. Further research is required to examine the inter-rater reliability and predictive accuracy of IPV risk assessment by threat assessors themselves, because they are using these tools in the field to make potentially high stakes risk management recommendations based on their results. Our inter-rater reliability sample size was 32, which is similar to or larger than the samples most studies in this field have used (Gerbrandij et al., 2018; Grann & Wedin, 2000; Hilton & Eke, 2016; Hilton et al., 2004; Hilton, Harris, Popham, et al., 2010; Jung & Buro, 2017; Rettenberger & Eher, 2013; Thijssen & de Ruiter, 2011), with the exception of Kropp and Hart (2000). A larger sample in future studies would enable tests of inter-rater reliability in subgroups, such as cases of IPV perpetrated by individuals of Indigenous descent.

The B-SAFER recent subscale is intended to be scored based on information present in the four weeks prior to the assessment. In our study, this time period fell in between the index offense and the completion of the threat assessment report, so there was minimal information relevant to recent events in the case files. It is also possible that, despite our coders being trained, periodically reviewed, and consistently referred to the detailed coding manuals, this time distinction was confusing and led to unreliable coding. Future research scoring the B-SAFER from archival records may require careful attention to the operationalization of the four-week time period.

We were unable to examine inter-rater reliability of the summary risk ratings of the SARA, SARA-V3, and B-SAFER because our research coders did not make final risk ratings. We have begun to study IPV risk assessment by the threat assessors themselves, and our initial examination of their final reports indicates that they rated risk as high in almost all cases in our sample. Given that the developers of structured professional tools advocate for decisions to be based on summary risk ratings rather than on mechanically tabulated total scores (e.g., Kropp & Hart, 2015), future research should be conducted with larger samples that include sufficient numbers of cases rated at lower levels of risk to permit informative tests of reliability. It would also be of interest to examine the concordance of summary risk ratings on these tools with the numeric risk categories on the ODARA (which has no verbal risk categories such as 'high risk' or 'low risk'). Given that threat assessors may use multiple tools to assess IPV risk (as they do in the threat assessment service in our study), it would be valuable to understand how they combine information from multiple tools to aid their summary risk ratings and inform their risk management recommendations.

A substantial portion of our sample (24%) was identified as of Indigenous heritage. The tools we used in this study have not been specifically validated for this population, although one published study found no significant difference between the ODARA's predictive accuracy for Indigenous and non-Indigenous men in a Canadian correctional sample (Hilton, Harris, Popham, et al., 2010). The appropriateness of threat assessors using existing IPV risk assessment tools for this population could be subject to debate, and future research should examine the predictive validity and cultural safety of IPV risk assessment for individuals of Indigenous heritage.

Implications for threat assessment practice

Because extensive professional training for threat assessors is expensive and time consuming, threat assessment is a limited and valuable service that must be reserved for the highest risk cases. Our findings suggest that investigating police officers effectively identify relatively high risk cases that warrant the allocation of additional risk management resources in the form of formal threat assessment consultation.

Given our observation that most of the referred cases were ultimately rated as high risk by threat assessors, it is fair to question the added value of threat assessment. The Risk, Need, Responsivity (RNR) principles of effective correctional service (Bonta & Andrews, 2017) stipulate that resources should be allocated in accordance with risk, such that individuals at higher risk of reoffending receive more risk management and other interventions. Our findings suggest that threat assessment services are indeed allocated to the highest risk cases. Furthermore, the threat assessor's role is not to simply generate a score, ranking, or summary judgement classifying the risk of future violence, but also to provide recommendations for a comprehensive risk management strategy. On the basis of their specialized training and experience, threat assessors' reports and testimony are routinely admitted as expert witness evidence in criminal and family courts. Consequently, threat assessors' risk-related opinions are likely to be given consideration and weight by judges making decisions about imposing risk-mitigating restrictions on at-risk individuals. As these decisions can take some time to be established, especially given threat assessment caseloads, it would be beneficial to explore whether threat assessment services can offer interim risk management suggestions for the referring police services to enact. In addition, the benefits of using a more extensive system of summary risk ratings in the threat assessment setting could be considered, such as the proposed standardized risk categories that include five levels rather than three (e.g., Hanson et al., 2017).

Conclusion

This study is the first investigation of the reliability of risk assessment when scored from the police and criminal records of a population of individuals referred for threat assessment following IPV offenses. We found that IPV cases referred to a police threat assessment service were measurably higher risk, as assessed by the SARA, B-SAFER, and ODARA, than routine samples of individuals apprehended by police for IPV perpetration. Our findings suggest that referrals from frontline police investigators are appropriate and consistent with the risk principle of effective correctional practice, in which the highest risk cases should receive the most intensive intervention (e.g., Bonta & Andrews, 2017). Furthermore, our findings indicate that the overall reliability of these measures within this relatively high risk sample was good, in terms of both internal consistency and inter-rater reliability for the total scores of the four IPV risk assessment tools, and most subscale scores. These results help to establish an important and necessary first step in evaluating the utility and effectiveness of threat assessment services, and in particular the appropriateness of referrals. Future research is needed to evaluate their predictive validity among men referred for IPV threat assessment.

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Dr. Hilton is an author of the ODARA and declares a financial interest in a publication cited in this article. No potential conflict of interest was reported by the other authors.

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